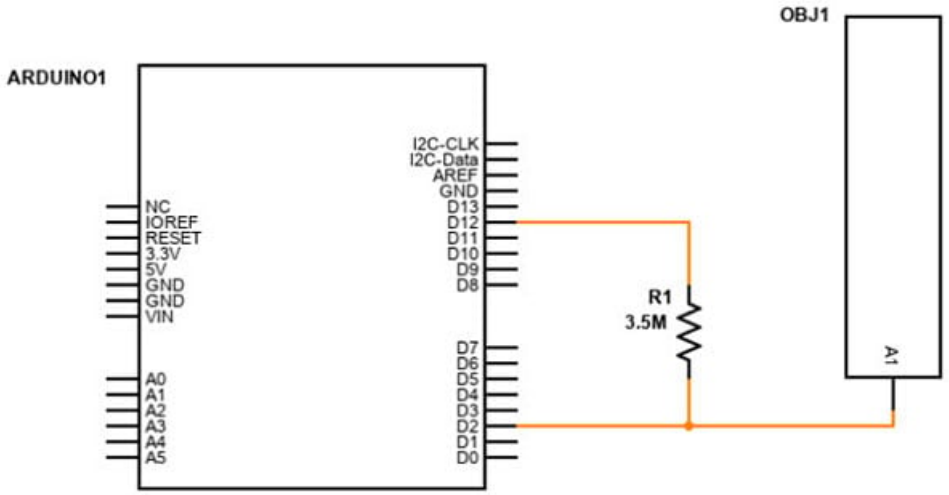


The software part of this project uses an Arduino library that sends out pulses and measures how long it takes to receive an answer. The library takes care of normalizing the result, and it returns an arbitrary number that describes the sensed capacitance. The sketch for this example only contains a few lines of code:

Copy Code

```
#include <CapacitiveSensor.h>

CapacitiveSensor sensor = CapacitiveSensor(2,12);

void setup()
{
  Serial.begin(9600);

  // Disable the automatic re-calibration feature of the
  // capacitive sensor library
  sensor.set_CS_AutocaL_Millis(0xFFFFFFFF);
}

void loop()
{
  long current_millis = millis();
  long capacitance = sensor.capacitiveSensor(30);

  // Print the result of the sensor reading
  // Note that the capacitance value is an arbitrary number
  // See: https://playground.arduino.cc/Main/CapacitiveSensor/ for details
  Serial.println(capacitance);

  // Wait for 50 milliseconds
  while(millis() - current_millis < 50);
}
```

The setup method turns on the serial port for debugging, and it then deactivates the automatic re-calibration feature of the CapacitiveSensor library. Please review the library's documentation for further details.

The loop method uses the library's capacitiveSensor function to obtain a capacitance reading. Note that the result is an arbitrary value that correlates to how close a user comes to the metal object. The loop method then prints that value to the serial console before it repeats these steps after a 50-millisecond delay.

The Arduino IDE's serial plotter is a useful tool for visualizing the output: